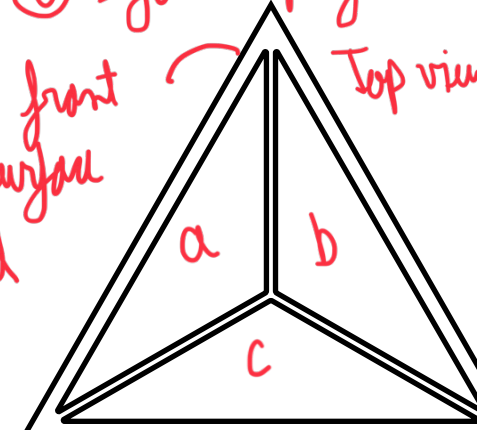


$E[\text{Steps}] = \frac{1}{\frac{1}{4}} = 4$
 via $\square$
 Can this be concluded as
 Andy will know if
 n.o of Steps > 4
 $\therefore P = P(X > 4)$
 $= 1 - P(X \leq 4)$
 $= \frac{48}{81}$ By Algorithm

$\square$ The Algorithm needs
 to rewritten
 1) The Sphere also needs to be represented
 Since 4 vertices every pair joined
 we are gonna call it K4
 & Honeycomb he
 2) We need to keep track of L R B
 for which we need to know from
 which edge the previous move came
 also for **he** we need to order the
 mbs into a CCW order so that we
 can determine L R B from arrived
 from

3) Also BFS needs to be joint BFS
 as tracking moves on both world
 together.
 $\odot \rightarrow$ for keeping track on K4

 $\therefore d$ has CCW
 mbs
 as c a b
 $a \rightarrow c \ b \ d$
 $b \rightarrow d \ a \ c$
 $c \rightarrow b \ a \ d$
 But these
 in a
 dict
 & now L R B can be determined
 if we know which face arrived from
 from

ANS $\rightarrow$ 0.55

Andy's Afternoon Amble



Andy the ant has moved on from his classic "Telstar" soccer ball homeland to live on a simpler spherical surface consisting of four white hexagons that are surrounded by alternating black triangles and white hexagons (three of each), and four black triangles surrounded by three white hexagons. To us this land is a truncated tetrahedron blown up into a sphere we see above on the left. Due to Andy's tiny size and terrible eyesight, he doesn't notice the curvature of the land and avoids the black triangles because he suspects they may be bottomless pits.

Much like his morning routine, every afternoon he wakes up from his nap on a white hexagon, leaves some pheromones to mark it as his special home space, and starts his random amble. Every step on this walk takes him to one of the three neighboring white hexagons with equal probability. He ends his amble as soon as he first returns to his home space, which he recognizes but cannot distinguish the edges of (i.e. he doesn't know if he returned across the same edge as he left). As an example, on exactly 1/3 of afternoons Andy's amble is 2 steps long, as he randomly visits one of the three neighbors, and then has a 1/3 probability of returning immediately to the home hexagon.

This afternoon his truncated tetrahedral homeland bounced through the very same kitchen with an infinite regular hexagonal floor tiling consisting of black and white hexagons, shown above on the right. In this tiling every white hexagon is surrounded by alternating black and white hexagons, and black hexagons are surrounded by six white hexagons. Andy fell off the ball and woke up on a white hexagon. He didn't notice any change in his surroundings, and goes about his normal amble.

Throughout his walk, Andy remembers the turns he's taken. Let p be the probability that by the end of his afternoon amble on this new land he has discovered that he is no longer on the truncated tetrahedral sphere. Find p in exact terms.

Can the Same Reasoning be applied here
 Am I missing sth?

Does this mean he remember n.o of steps if
 yes 48/81 correct
 OR Sth Else? // Because otherwise
 quite an easy
 puzzle.

Or is it like he
 remembers left right
 as if he takes 3 left / right on the
 Sphere he reaches home but not here
 $\rightarrow$ Also on his sphere he can reach home
 in odd steps but not here

Actually you know what this is a different
 puzzle here the key is any seq of moves
 $\{L, R, B, \dots\}$ which should have
 gotten him home on his spherical
 planet down if got him here, then he
 knows //

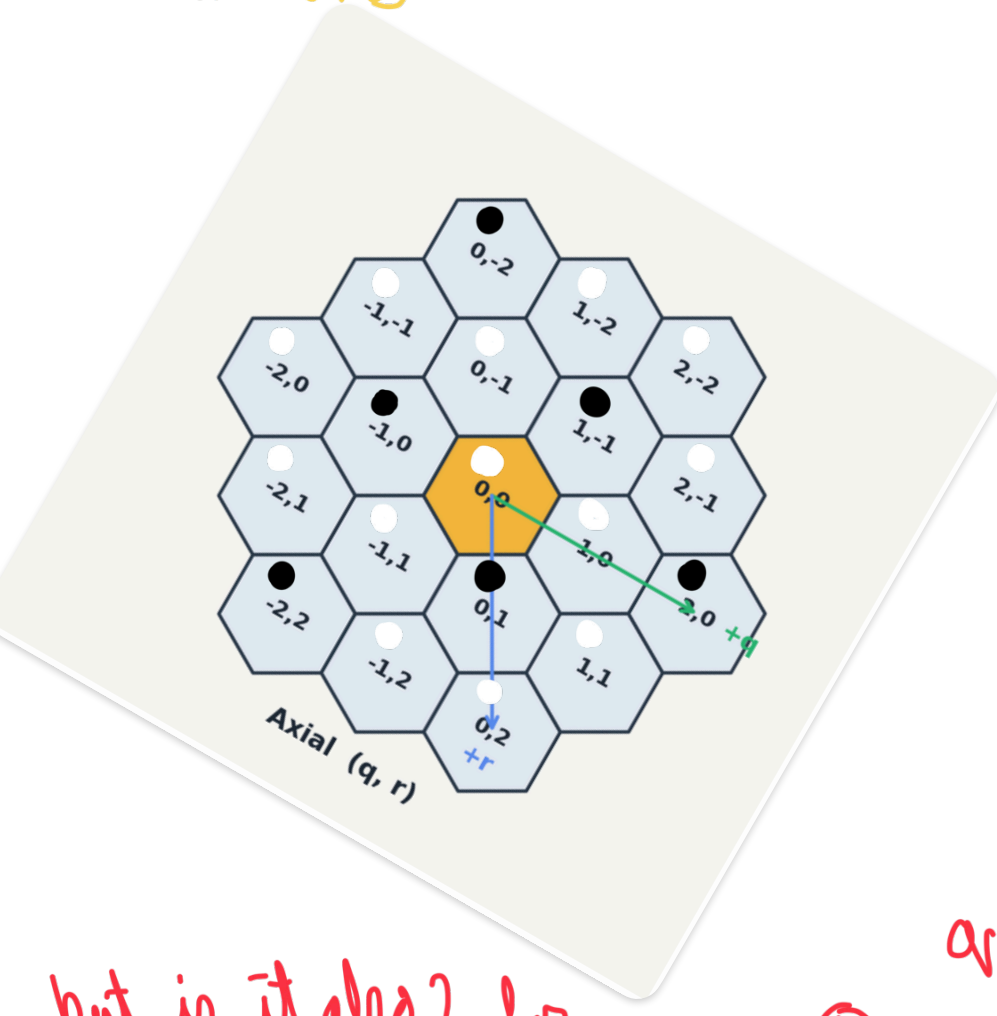
So for a sequence of turns $\{L, R, L, R, R, B, \dots\}$
 if H = The step on which he actually gets home
 & H' = The step on which he would have gotten
 home if he was on the sphere

then prob he knows = $P(H \neq H')$
 always Even can be odd

Can be represented
 in a +ve recurrent
 Markov Chain
 $\therefore$ By

$\square$ Kac's Return Time
 Formula
 $E[\text{steps}] = \frac{1}{\pi_i} = \frac{1}{\frac{1}{20}} = 20$

For Algorithm we need a
 way to represent the grid.)
 Quick Google Search suggests
 using Axial Words
 After learning " " the representation goes
 sth like this.

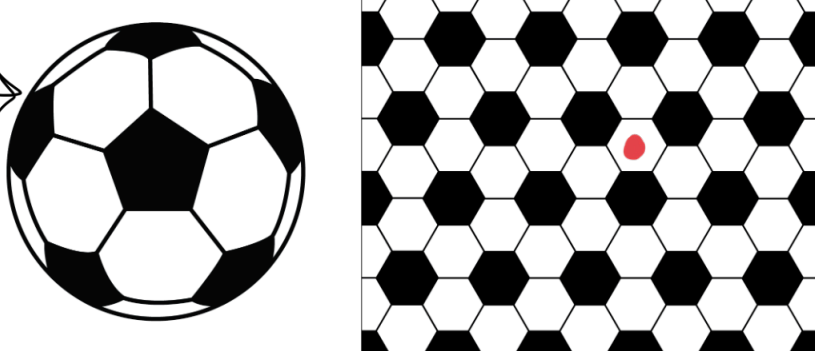


but is it also 2 for
 white seq
 $(q-r) \bmod 3$
 $0 \ 1 \ 0 \ -2=1 \ 0$
 $1 \ 0 \ -2=1 \ 0$
 $0 \ 1 \ 0 \ -2=1$
 $\therefore$ white seq 0 & 1 $\rightarrow \triangle$

Andy's Morning Stroll

JULY 2022 : PUZZLE

Show Solution



Andy the ant has spent most of his days living on a strange land consisting of white hexagons that are surrounded by alternating black pentagons and white hexagons (three of each), and black pentagons surrounded by five white hexagons. To us this land is familiar as the classic soccer ball we see above on the left. Due to Andy's tiny size and terrible eyesight, he doesn't notice the curvature of the land and avoids the black pentagons because he suspects they may be bottomless pits.

Every morning he wakes up on a white hexagon, leaves some pheromones to mark it as his special home space, and starts his random morning stroll. Every step on this stroll takes him to one of the three neighboring white hexagons with equal probability. He ends his stroll as soon as he first returns to his home space. As an example, on exactly 1/3 of mornings Andy's stroll is 2 steps long, as he randomly visits one of the three neighbors, and then has a 1/3 probability of returning immediately to the home hexagon.

This morning, his soccer ball bounced through a kitchen with an infinite (at least practically speaking...) regular hexagonal floor tiling consisting of black and white hexagons, a small part of which is shown above on the right. In this tiling every white hexagon is surrounded by alternating black and white hexagons, and black hexagons are surrounded by six white hexagons. Andy fell off the ball and woke up on a white hexagon. He didn't notice any change in his surroundings, and goes about his normal morning routine.

Let p be the probability that his morning stroll on this new land is strictly more steps than the expected number of steps his strolls on the soccer ball took. Find p , rounded to seven significant digits.

For the ∞ Regular Hexagonally Tiled Floor $\rightarrow$ Markov Chain is
 Null Recurrent : No use

$P(X > 20) = 1 - P(X \leq 20)$
 $P(2) = 1/3 = 3/9$
 $P(3) = 0 = 0/27$
 $P(4) = 6/81$
 $P(5) = 0/243$
 $P(6)$

1 Loop
 3 Black Hexagons
 $\therefore 3 + 1 \ 2 = 15/729$
 Worst X
 $\rightarrow \triangle$ also a
 Possibility like
 $\therefore 3 \times 2 = 6$

Searching
 Pattern
 for Black
 Hexagons
 $q+r$
 -2
 0
 2
 1
 0
 $q-r$
 2
 2
 2
 -1
 -1
 -4
 Nothing
 obvious
 if mod 3
 $\therefore (q-r) \bmod 3$
 $= 2$
 for black
 squares

ANS $\rightarrow$ 0.4480325561718

If you go & retrase
 its always 2 n
 steps
 Special case could have been
 a loop around hexagon
 but 6 $\rightarrow$ 10 $\rightarrow$ 14 steps
 1 Hex 2 Hex 3 Hex
 It seems loop around
 n Hexagon = $2 + 4n$
 $\therefore$ Not Possible to return
 in odd moves

$\triangle \rightarrow$ White Neighbours
 if $(q-r) \bmod 3 = 0$
 $\rightarrow (q+1, r); (q, r-1); (q-1, r+1)$
 if $(q-r) \bmod 3 = 1$
 $\rightarrow (q-1, r); (q+1, r-1); (q, r+1)$